

AI-Based Fake Identity & Document Screening System

System Architecture, Document Scope & Hackathon Winning Blueprint

Problem ID: 26188 Ministry: Ministry of Home Affairs (MHA)

Agency: Sashastra Seema Bal (SSB), Police II Division

Theme: Blockchain & Cybersecurity

Executive Analysis: Understanding SIH Problem Statement 26188

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- **Title:** AI-Based Fake Identity & Document Screening System
- **Ministry & Department:** Ministry of Home Affairs (MHA) — Sashastra Seema Bal (SSB), Police II Division
- **Theme:** Blockchain & Cybersecurity
- **Category:** Software

The Operational Context: SSB Border Checkpoints

The client agency is **Sashastra Seema Bal (SSB)**. Unlike airport immigration counters (which handle mostly international ePassports), the SSB guards the **1,751 km Indo-Nepal border** and the **699 km Indo-Bhutan border**. These are open borders governed by historic bilateral treaties:

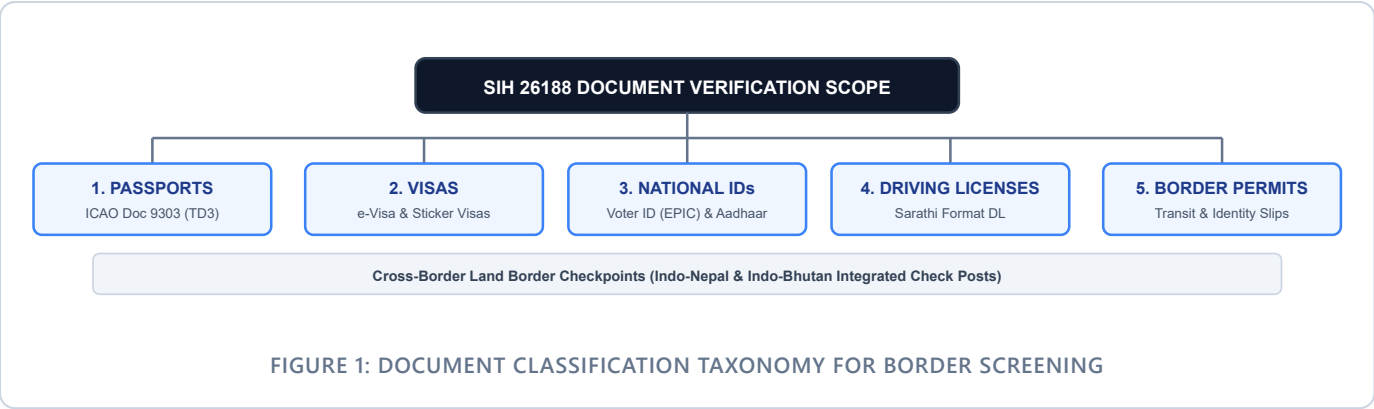
1. **Indian and Nepalese/Bhutanese citizens** do not require a passport or visa to cross overland Integrated Check Posts (ICPs like Raxaul, Sonauli, Jogbani, Panitanki). They travel using domestic identification documents: **Voter IDs (EPIC), Aadhaar cards, Driving Licenses, Embassy Identity Slips, and Border Transit Permits**.
2. **Third-country nationals (foreigners)** must present standard **Passports, valid Indian Visas (e-Visa or sticker), and physical border entry/exit ink stamps**.
3. **Core Threat Vectors:** Trans-border smuggling syndicates, illegal immigration, and cross-border operatives frequently exploit this open border using **counterfeit Voter IDs/Aadhaar cards, altered DOBs on permits, tampered immigration ink stamps, swapped portrait photos, and identity impersonation (using multiple identities under different names)**.
4. **Theme Alignment:** The problem is explicitly classified under **Blockchain & Cybersecurity**. Developing computer vision models alone will fail the evaluation criteria if the **immutable audit trail, zero-knowledge verification, and tamper-evident data custody** are missing.

Operational Imperative

At peak transit hours, border checkpoints process thousands of crossings. Manual verification causes severe congestion. The screening solution must execute across OCR, forensic tamper analysis, facial biometrics, and registry validation within **under 2.5 seconds**.

1. Documents Verification Scope: What Documents Must Be Supported?

To cover the operational mandate of the SSB and Ministry of Home Affairs, the screening system must handle five distinct document classes:



DOCUMENT CATEGORY	DOCUMENT TYPES	KEY FIELDS EXTRACTED	SPECIFIC VALIDATION LOGIC
1. International Passports	Standard TD3 Passports (36/60 pages)	Full Name, Passport No, Nationality, DOB, Expiry, Gender, Issuing State, MRZ Line 1 & Line 2.	- ICAO Doc 9303 repeating 7-3-1 check digit algorithms (Passport number, DOB, Expiry, Composite). - VIZ-to-MRZ cross-field consistency check. - ISO 3166-1 alpha-3 country code validation.
2. Visas & Entry Approvals	Indian Sticker Visas, e-Visa Grant Letters	Visa Number, Visa Type (Tourist, Business, Transit), Valid From, Valid Until, Number of Entries, Stay Duration, Linked Passport Number.	- Date bounds checking (<code>valid_until</code> $\geq$ <code>valid_from</code>). - Passport number cross-reference between visa and passport. - Allowed stay duration vs. calculated date window.
3. National Identity Cards	Voter ID (EPIC) & Aadhaar Card (Vital for Indo-Nepal/Bhutan transit)	Document ID, Full Name, Father's/Husband's Name, DOB / Year of Birth, Gender, Address.	Aadhaar: Verhoeff checksum algorithm on 12-digit UID. Voter ID: Election Commission regex (3 alpha prefix + 7 numeric digits, e.g. <code>ABC1234567</code>). - VIZ name vs. database query.
4. Driving Licenses	State Transport (Sarathi format DL)	DL Number, Holder Name, DOB, Issue Date, Expiry Date, Vehicle Class (LMV/MCWG).	- Regex structure check: 2-letter State Code + 2-digit RTO + 4-digit Year + 7-digit Serial Number. - Expiry validation against current date.
5. Border Transit Permits	Emergency Transit Slips, Embassy ID Slips, Overland Authorizations	Permit Reference ID, Designated Border Post (ICP), Valid Dates, Authorized Route.	- ICP authority signature/stamp check. - Non-extendable date expiration check.

2. The SIH 188 Winning Strategy: Why Teams Lose vs. How You Will Win

Judges from the Ministry of Home Affairs and border police divisions evaluate hundreds of presentations. The table below contrasts typical failed submissions with a winning approach:

EVALUATION DIMENSION	WHY TYPICAL TEAMS LOSE	HOW YOU WILL WIN (THE GOLD STANDARD)
Scope Understanding	They only demo an airport passport MRZ reader.	You demonstrate both international passports/visas AND Indian/Nepal border documents (Voter ID, DL, Transit Permits) tailored for the SSB .
"Multiple Identities" Requirement	They ignore it completely or treat it as a database text search.	You implement a 1:N Biometric Vector Search (FAISS) that instantly flags when the same physical face appears under two different names or document numbers.
Tampering Detection	They use only Error Level Analysis (ELA) or a generic CNN with no explainability.	You combine a multi-signal ensemble: Pixel ELA Heatmap + EXIF Metadata Tool Detection + Hugging Face ViT AI Forgery Classifier + Ink Stamp Hue/Boundary Analysis .
Blockchain Theme	They stick a non-functional Web3/Ethereum logo on their slide or do nothing.	You demonstrate a Permissioned Cryptographic Audit Ledger storing document hashes, officer IDs, and risk decisions in an immutable SHA-256 Merkle chain.
Explainable AI (XAI)	The system just outputs "Risk: 85% - FAKE" with zero explanation.	The system outputs granular reason codes : ["MRZ DOB Checksum Failed", "Canva signature in EXIF", "Face Match 0.38 < 0.60", "Aadhaar Verhoeff Failed"] .
Processing Speed	The system takes 20+ seconds per scan, which fails at high-traffic border posts.	Complete asynchronous pipeline processes a document in under 2.5 seconds , meeting high passenger volume operational needs.

3. Detailed System Architecture & Workflow Diagrams

End-to-End Operational Workflow at Border Checkpoint

The complete flow from traveler arrival at the SSB ICP to risk assessment and ledger logging:

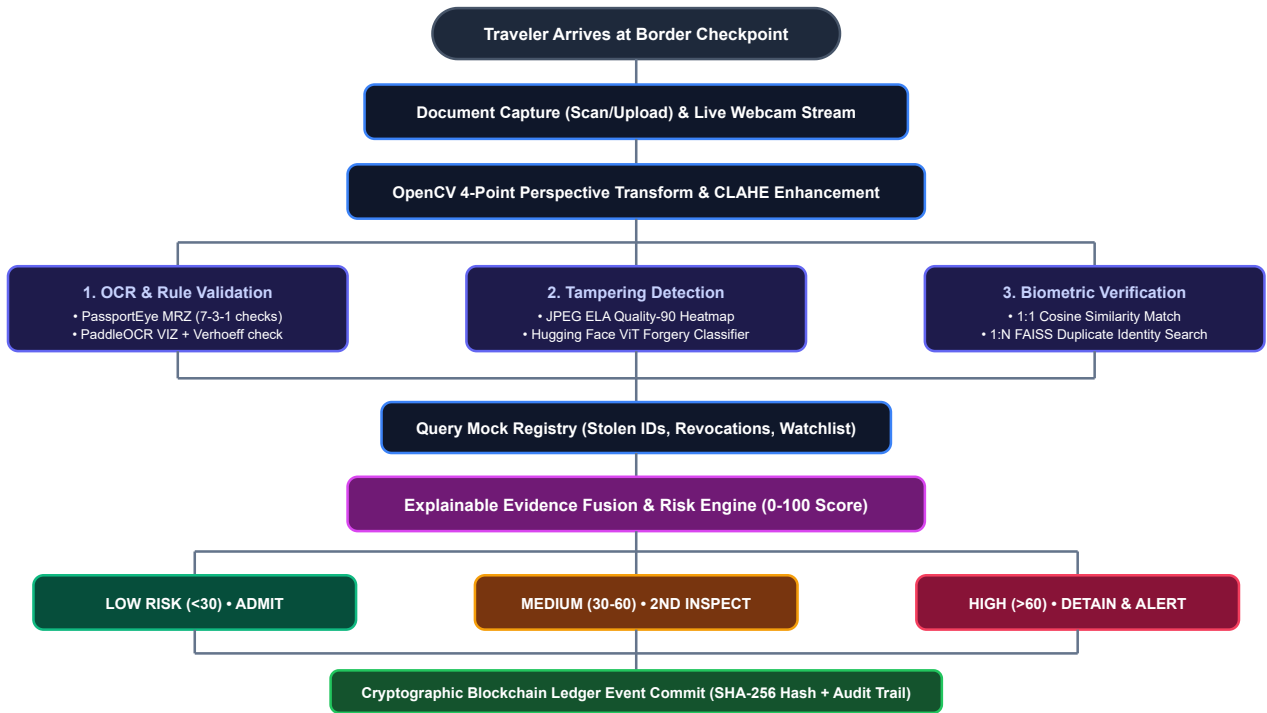
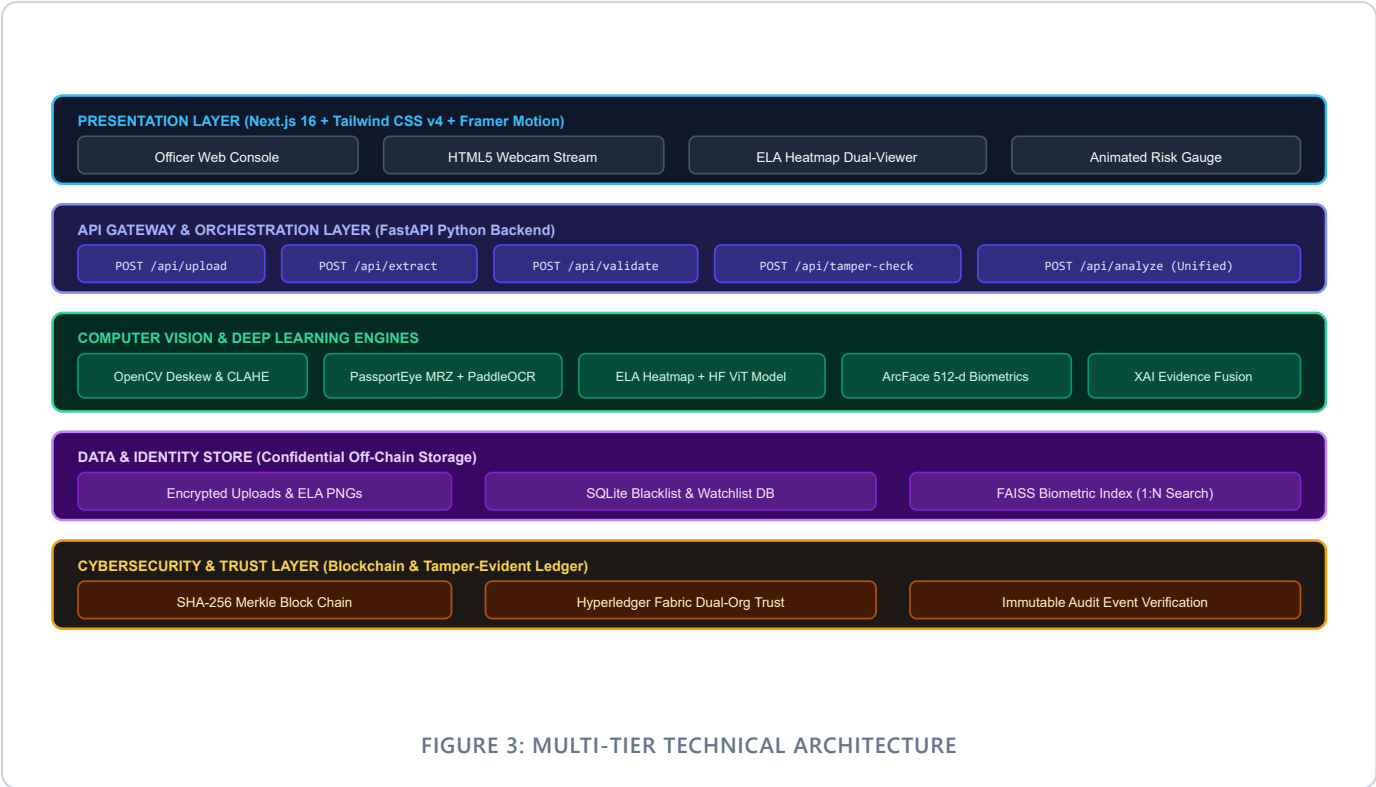


FIGURE 2: END-TO-END OPERATIONAL SCREENING WORKFLOW

Detailed System Technical Architecture

Layered micro-architecture connecting the Next.js frontend, FastAPI backend, AI models, and cryptographic ledger:



4. Deep Dive into the 6 Core Modules

Module 1: Universal OCR & Document Understanding

- Perspective Correction & Preprocessing:** Automatically rescales input images, computes Gaussian blur, detects quadrilateral document borders using `cv2.approxPolyDP`, and maps them to a top-down view with `cv2.warpPerspective`. CLAHE eliminates uneven illumination and checkpoint glare.
- Dual-Zone Extraction:**
 - MRZ Extraction:** Utilizes `PassportEye` and `Tesseract` with tight character whitelisting and scoped positional letter-to-digit auto-correction (e.g. `0` → `0`, `I` → `1`, `B` → `8` in numeric slices).
 - VIZ Extraction:** Utilizes `PaddleOCR` for free-text fields (holder name, address, father's name) and regex parser for Indian national documents (Voter ID, Aadhaar, Driving License).

Module 2: Document Validation (Rules & Standards)

- ICAO Doc 9303 Checksums:** Computes standard 7-3-1 modulo 10 check digits across passport number, DOB, expiry date, and composite line 2 check:

$$\text{Check Digit} = (\sum \text{Value}(c_i) \times w_{i \bmod 3}) \bmod 10, \text{ where } w \in \{7, 3, 1\}$$

- Aadhaar Verhoeff Algorithm:** Implements the dihedral group D_5 multiplication table to validate 12-digit Aadhaar UID numbers and detect transpositions.
- VIZ-to-MRZ Cross-Check:** Detects cases where text in the visual zone was altered (e.g. Photoshop name change) without updating the MRZ.

Module 3: Tampering Detection (Core AI Innovation)

- **Error Level Analysis (ELA):** Re-compresses the image at JPEG Quality 90 and calculates absolute pixel variance:

$$\text{Diff}(x,y) = | I_{\text{original}}(x,y) - I_{\text{resaved}}(x,y) |$$

Generates an amplified ELA heatmap where modified or spliced regions appear highlighted due to compression mismatch.

- **Pretrained ViT Forgery Classifier:** Deploys a Hugging Face Vision Transformer (`zodumair/document-forgery-detector`) over alpha-blended ELA images to output AI tampering confidence.
- **EXIF Metadata Inspection:** Scans metadata headers for graphic editing tools (Photoshop, GIMP, Canva, Snapseed) and flags stripped metadata.
- **Stamp & Seal Forensics:** Analyzes ink color distribution (HSV color clustering) and edge continuity of border entry/exit stamps.

Module 4: Biometric Face Verification & Multi-Identity Detection

- 1:1 Verification (Impersonation Defense):** Auto-crops the document portrait, captures a live face frame from the checkpoint webcam, computes 512-dimensional ArcFace embeddings, and evaluates cosine similarity:

$$\text{Cosine Similarity} = (u \cdot v) / (||u|| \ ||v||)$$

Threshold calibrated: ≥ 0.65 (Match), $0.50\text{--}0.64$ (Uncertain), < 0.50 (Mismatch / Impersonation).
- 1:N Duplicate Identity Detection (Winning Feature):** Every screened face is indexed in an in-memory **FAISS** vector database. If a live face matches an existing record with similarity > 0.70 under a *different name or passport number*, a **"CRITICAL: Duplicate Identity / Alias Detected"** security alarm is raised immediately.

Module 5: Explainable Evidence Fusion & Risk-Scoring Engine

Combines multi-modal signals into a calibrated composite risk score between 0.0 and 100.0:

$$\text{Overall Risk Score} = \min(100, \text{Base Tampering Score} + \sum \text{Penalties})$$

TRIGGERED SIGNAL	PENALTY POINTS	DIAGNOSTIC REASON CODE
Blacklisted / Stolen Registry Match	+50.0 Points	"Passport blacklisted: Reported stolen"
Biometric Face Mismatch	+30.0 Points	"Biometric mismatch: similarity below 0.50"
MRZ Checksum Failure	+25.0 Points	"MRZ check digit mismatch on field(s): DOB, Composite"
Document Expired	+20.0 Points	"Document expired on 2024-02-14"
EXIF Editing Software Detected	+15.0 Points	"Editing software signature detected: Adobe Photoshop"
VIZ-to-MRZ Inconsistency	+15.0 Points	"Visual name does not match MRZ encoded name"
AI Forgery Model Alert	+20.0 Points	"AI ViT detector: High probability of digital forgery"

Module 6: Blockchain & Cybersecurity Trust Layer

- Zero Biometrics On-Chain:** Complies with DPDP Act. Never stores names, photos, or raw biometric arrays on-chain.
- Cryptographic Block Payload:**

$$\text{Block} = \{ \text{Index}, \text{Timestamp}, \text{PrevHash}, \text{SHA-256}(\text{Doc}), \text{RiskScore}, \text{OfficerID}, \text{Nonce} \}$$
- Dual-Authority Architecture:** Simulates Issuing Authority (Org 1) and Border Screening Authority (Org 2) with immutable append-only verification.

5. Production vs. Hackathon Prototype Feasibility

SYSTEM COMPONENT	PRODUCTION VISION (REAL WORLD)	HACKATHON WINNING IMPLEMENTATION (IN REPO)
Document Input	Specialized 3M/Thales multi-spectral border document scanners (UV, White, IR illumination).	Smartphone camera upload + top-down webcam capture + OpenCV perspective deskewing.
ePassport Chip	Physical ISO 14443 NFC smartcard reader querying ICAO PKD CSCA certificates.	Simulated Cryptographic LDS Validator (decodes mock signed payload and validates hash integrity).
National Databases	Live CCTNS, IVFRT, Interpol SLTD, and UIDAI e-KYC restricted APIs.	Curated SQLite Mock Registry (<code>blacklist.db</code>) with active, expired, stolen, and revoked test cases.
Face Biometrics	NIST FRTE-compliant certified biometric turnstile camera with active 3D IR liveness.	Live WebRTC webcam stream + ArcFace deep embeddings + 1:N FAISS vector search.
Blockchain	Multi-node Hyperledger Fabric network with raft ordering service across MHA data centers.	Cryptographic Tamper-Evident SHA-256 Merkle Ledger with verification API and chaincode contracts.

6. The 10 Live Demonstration Test Scenarios

Curated test matrix designed to demonstrate all functional requirements during the live judging round:

#	TEST CASE SCENARIO	TRIGGERED MODULE SIGNALS	EXPECTED OUTCOME
1	Genuine Clean Passport	All 7-3-1 check digits valid, VIZ matches MRZ, face match 0.88, no ELA anomalies.	LOW (Score: 8) • PASS
2	Expired Travel Document	Date validator flags expiration; checksums valid; face matches.	MEDIUM (Score: 48) • 2ND
3	Altered Date of Birth	MRZ DOB check digit fails (computed ≠ expected); ELA shows local variance.	HIGH (Score: 78) • REVIEW
4	Edited Name in VIZ	Printed name altered in VIZ; VIZ-to-MRZ cross-check flags discrepancy.	HIGH (Score: 74) • REVIEW
5	Digital Forgery (Canva/PS)	EXIF metadata reveals editing software; ViT AI classifier flags forgery probability > 80%.	HIGH (Score: 84) • REVIEW
6	Swapped Portrait Photo	Document face does not match live webcam; cosine similarity 0.32 < 0.50 threshold.	HIGH (Score: 88) • REVIEW
7	Multiple Identities (1:N)	Live face matches an existing traveler in FAISS index under a different name/passport.	HIGH (Score: 96) • ALARM
8	Blacklisted Stolen Passport	Passport number matched in SQLite stolen travel document database.	HIGH (Score: 98) • DETAIN
9	Invalid Aadhaar Number	12-digit Aadhaar UID fails mathematical Verhoeff checksum algorithm.	HIGH (Score: 76) • REJECT
10	Tampered Checkpoint Stamp	Visa stamp color histogram and boundary edge continuity fail authenticity bounds.	MEDIUM (Score: 56) • 2ND

7. Implementation Roadmap: What to Build Next

- Module 4 (Biometric Face Verification & 1:N FAISS Duplicate Search):**
 - Integrate document portrait auto-cropping using OpenCV / Haar cascade.
 - Add HTML5 webcam video capture component in Next.js frontend.
 - Implement FAISS vector index in FastAPI backend to detect duplicate travelers.
- Module 2 Addition (VIZ-to-MRZ Cross-Check & National ID Validation):**
 - Cross-verify visual OCR text against MRZ fields.
 - Implement Aadhaar Verhoeff checksum and Voter ID regex verification.
- Module 6 (Cryptographic Audit Ledger):**
 - Build immutable SHA-256 block ledger in backend with `/api/ledger/verify` endpoint.
 - Add verification badge and digital receipt drawer in Next.js UI.
- 10-Case Test Suite:** Assemble high-resolution synthetic test documents corresponding to the 10 demonstration scenarios.